

Algorithms

Fall 2026

General

- Algorithms refers to the control aspect of robotics
 - How a robot processes the input to solve a problem
- Since the “brain” of a robot is a computer, every computer science algorithm can be run on a robot
- Most of the time, however, we have limited processing power and memory onboard the robot’s computer

Algorithms as a Study

- Taught in the following classes:
 - 15-210, 15-251, 15-451, etc.
- Not necessarily relate to robotics problems, but are still applicable
 - Many 16-xxx (Robotics Institute) courses will teach more applied ones
- Huge field of study and extremely convoluted
- Let's not get into the specifics...

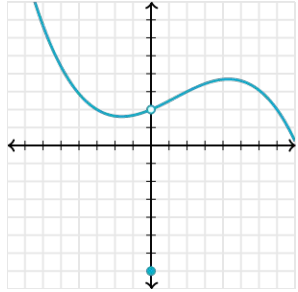
Today's Topic: Graphs

- Converting the real world to something more manageable (read: can be asked on LeetCode):
 - Abstraction of problems and namely graphs!!!!!!!
- Several types of algorithms depending on the environment:
 - Informed: You know where you are and where to go
 - Uninformed: Everything around you is a mystery
- **Path planning**: wanting to go from A to B
- Also called **graph search** (Searching the graph for the goal)

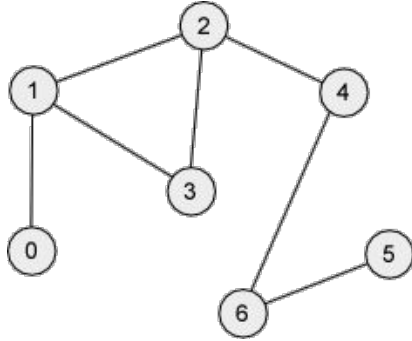
Important Questions

- Why do we need to change real world problems to simple ones?
- How do we choose the way we represent a problem?
- What do we need to retain from a real world problem to a simplified one?
- When do we know when to stop simplifying a problem?

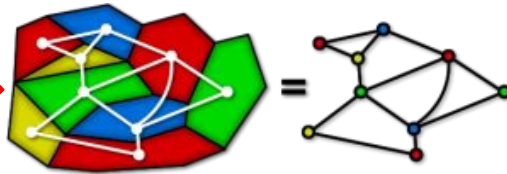
Graphs



Going from
this idea...

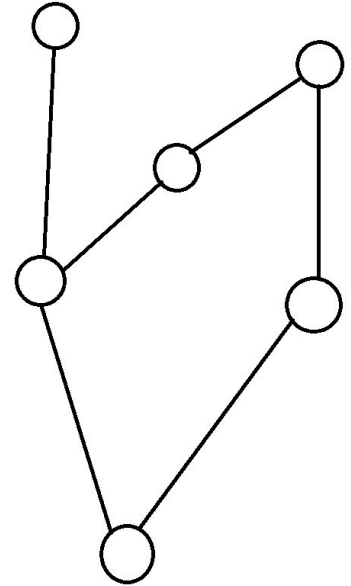
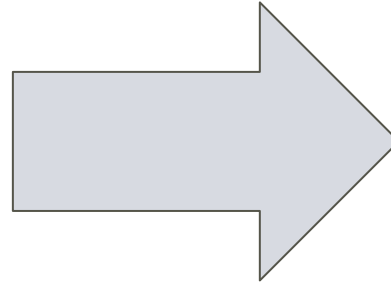
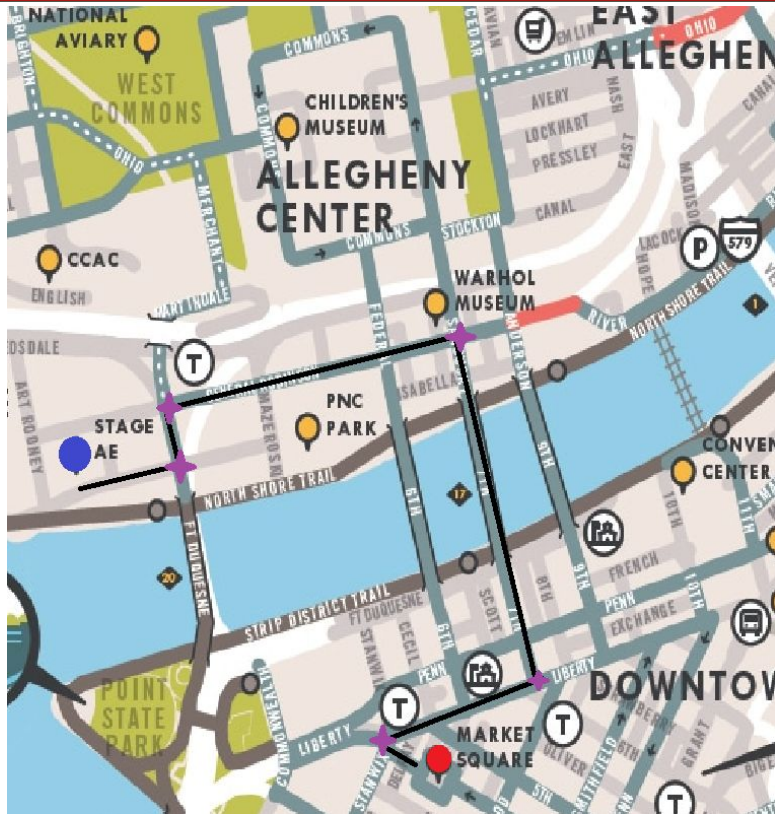


To this idea!



- Graphs are described as a set of points (vertices) connected by lines (edges)
- Due to their simplistic design, they can be used to model a variety of real world environments
- Each vertex can be connected to any number of vertices in the graph through an edge (even itself).
- We use graphs to simplify the real world (a room, a city) and find paths for our robots

Moving Through a City

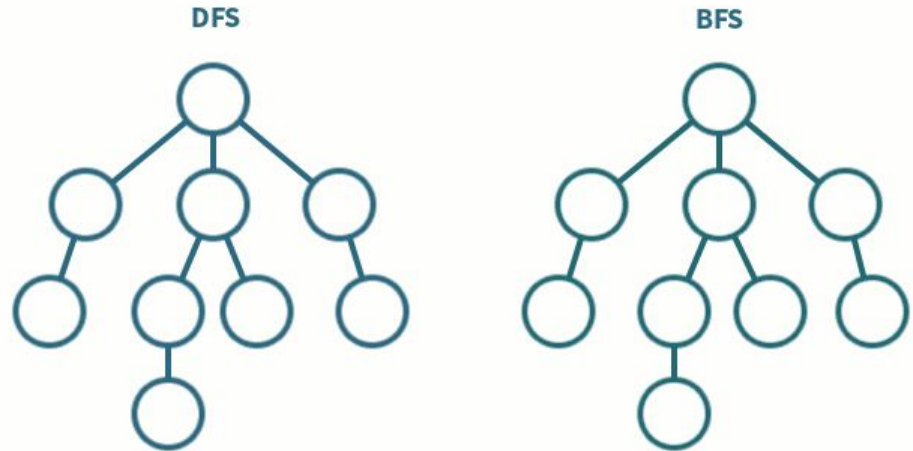


Important Answers

- Why do we need to change real world problems to simple ones?
 - Solving simple problems helps solve more complicated ones
- How do we choose the way we represent a problem?
 - Look at what's needed to solve it
- What do we need to retain from a real world problem to a simplified one?
 - Basic characteristics (variables) and their relationship
- When do we know when to stop simplifying a problem?
 - When doing so changes the nature of the problem or its variables

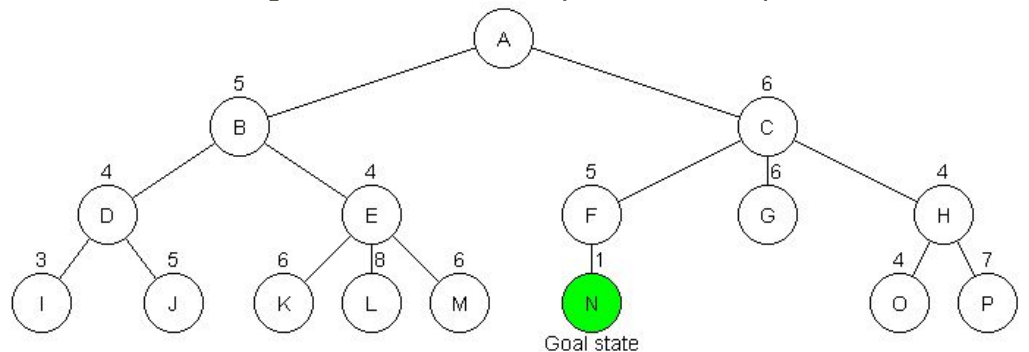
Uninformed: BFS and DFS

- **DFS (Depth First Search)**
follows a path to its end, and then selects the next possible path by backtracking
- **BFS (Breadth First Search)**
covers all the closest points available and then moves to the next depth
- They have the same average run time

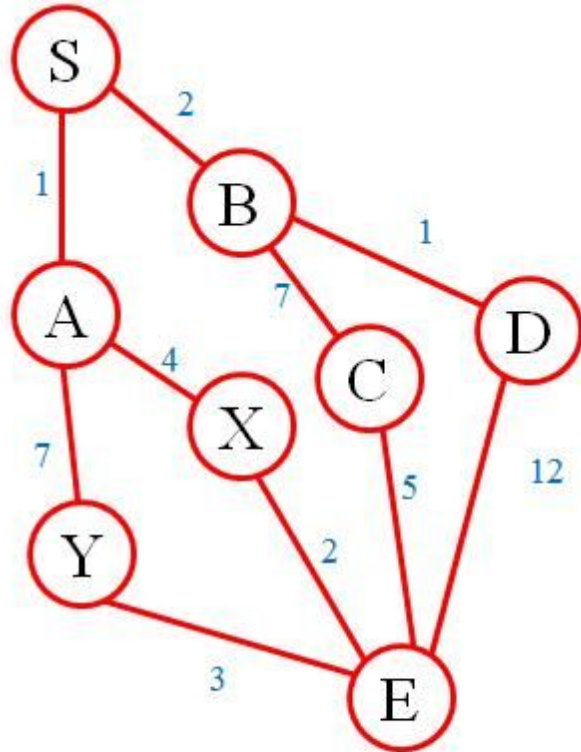


Informed: A* search

- There are cases that you would rather move through a specific path to a point rather than take another (rubble, fire, etc).
- We can model that behaviour by weighing our path and searching through A*.
- A* just looks at every available path and chooses the best one based on a set of rules you set for it (heuristic).



how2examples.com



■ Values for h:

A:5, B:6, C:4, D:15, X:5, Y:8

Expand S

$$\{S,A\} f=1+5=6$$

$$\{S,B\} f=2+6=8$$

Expand A

$$\{S,B\} f=2+6=8$$

$$\{S,A,X\} f=(1+4)+5=10$$

$$\{S,A,Y\} f=(1+7)+8=16$$

Expand B

$$\{S,A,X\} f=(1+4)+5=10$$

$$\{S,B,C\} f=(2+7)+4=13$$

$$\{S,A,Y\} f=(1+7)+8=16$$

$$\{S,B,D\} f=(2+1)+15=18$$

Expand X

$\{S,A,X,E\}$ is the best path... (costing 7)

A* Candy Question

- A* is faster, on average, than uninformed searches, and takes into account information about the environment.
 - Why not just use A* for everything?



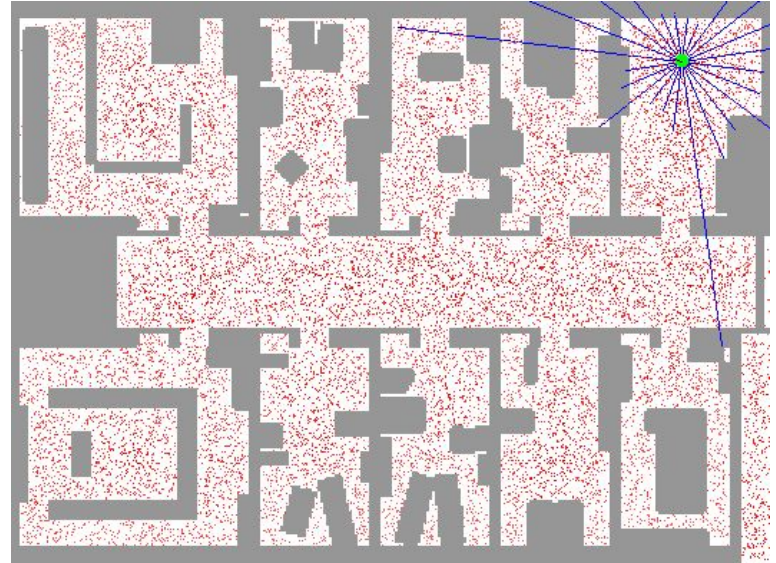
A* Candy Answer!

- A* requires information about the environment and processing power
 - What if you enter a cave with no map?
 - What if you don't have the processing power?



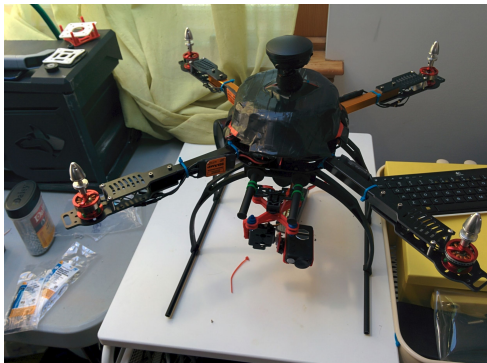
Localization

- Sometimes you don't know where you are in an environment
- You can either know the map and not know your location in it, or have no idea about your surroundings.
- In the former case, you can determine where you are by moving
- This process is called localization



Putting it all together: SLAM

- SLAM lets robots make a map of an unknown environment while keeping track of where they are within that map
- Used to map your environment in relation to self to make a map of your environment as you move



Slam & control in action

